

MATH 3310 HOMEWORK ASSIGNMENT 6

DUE ON THURSDAY 13 OCTOBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

(1) Prove that an integer n is odd if and only if n^2 is odd.

(2) Prove the following statement.

Let $m, n \in \mathbb{Z}$. The product mn is odd if and only if m and n are both odd.

(3) Consider the following argument:

Let n be an even number and write $n = 2k$ for some $k \in \mathbb{Z}$.

One then has

$$n^2 - n = (2k)^2 - 2k = 2(2k^2 - k),$$

and as $2k^2 - k$ is an integer, $n^2 - n$ is even.

For each of the following statements, decide if it is proved by the argument and explain why/why not?

- (a) If $n^2 - n$ is odd, then n is odd.
- (b) If n is odd, then $n^2 - n$ is odd.
- (c) For every integer n , the number $n^2 - n$ is even.
- (d) If n is even, then $n^2 - n$ is even.

(4) Prove the following statement:

For every integer n the integer $n^2 + n + 7$ is odd.

(5) Negate each of the following statements and decide in each case if it is the original statement or the negation that is true.

- (a) $\forall x \in \mathbb{R}, x^2 > 0$
- (b) $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}, |x + y| < |x| + |y|$.
- (c) $\exists n \in \mathbb{Z}, n + n = n^n$.
- (d) $\exists p \in \mathbb{R} - \mathbb{Q}, \exists q \in \mathbb{Q}, p^q \in \mathbb{Q}$.